AWS API Gateway: API Keys and Usage Plans

To Begin with the Lab

Summary of the Lab

In this lab, an **API key** and a **usage plan** were configured for an existing **REST API** in **API Gateway**. The API already had methods deployed to a stage, so the setup focused on controlling access rather than rebuilding the API itself. A key named user-api-key was created, a usage plan named basic-usage-plan was created, and the REST API stage Dev was attached to that plan. The API key was then attached to the usage plan, and the POST method was marked as **API key required** before redeploying the API. The lab confirmed the behavior in **Postman**: without the x-api-key header the request returned **403 Forbidden**, and with the header the request reached **Lambda** successfully. Later, enabling quota caused requests beyond the allowed limit to return **429 Too Many Requests**.

Understanding API Keys and Usage Plans

API keys and **usage plans** exist so you can control and track how clients use your **API Gateway REST API**. The API key identifies the client, while the usage plan defines which API stage the key can access and can also enforce throttling or quotas. This helps protect the backend, limit abuse, and manage access at the client level. In this lab, the method-level setting **API Key Required = True** made the POST route reject requests that did not include the required header.
Example:
A client sends a POST /user request to create a new user record. If the request does not include x-api-key, API Gateway blocks it with **403 Forbidden**. When the same request includes the valid key, the request is accepted and reaches the Lambda function that writes the record to DynamoDB.

Lab Steps

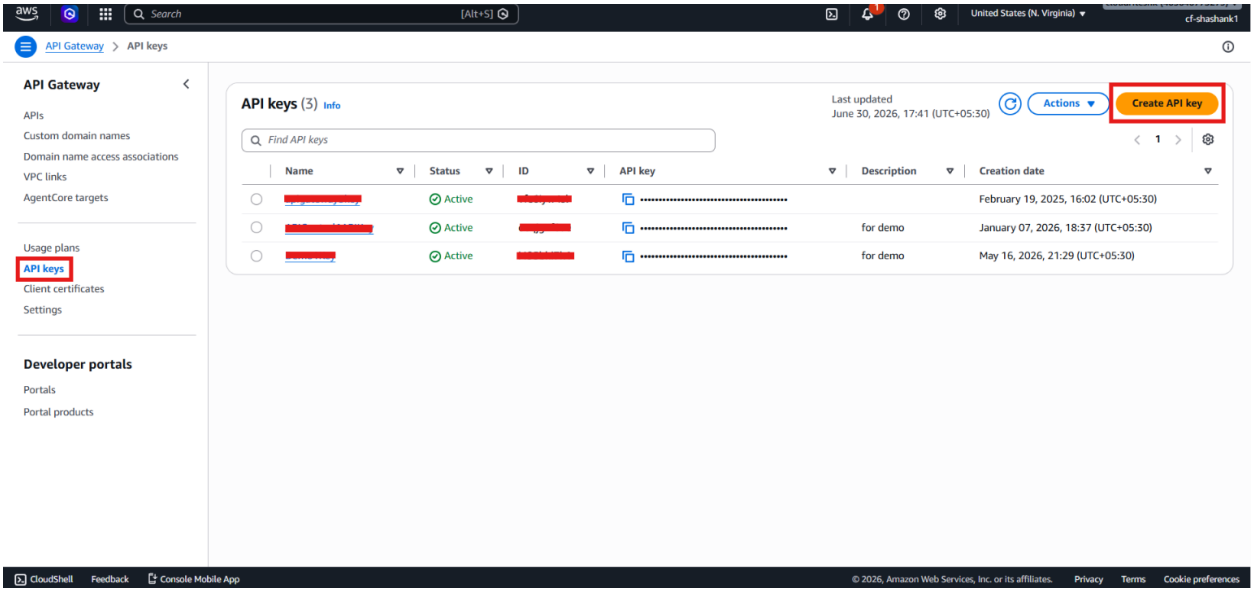
Step 1 – Confirm the Prerequisites

- Sign in to the AWS Management Console.
- Open **API Gateway**.
- Confirm that the **REST API** already exists.
- Confirm that the methods (GET, POST, DELETE) are already deployed to a stage such as dev.

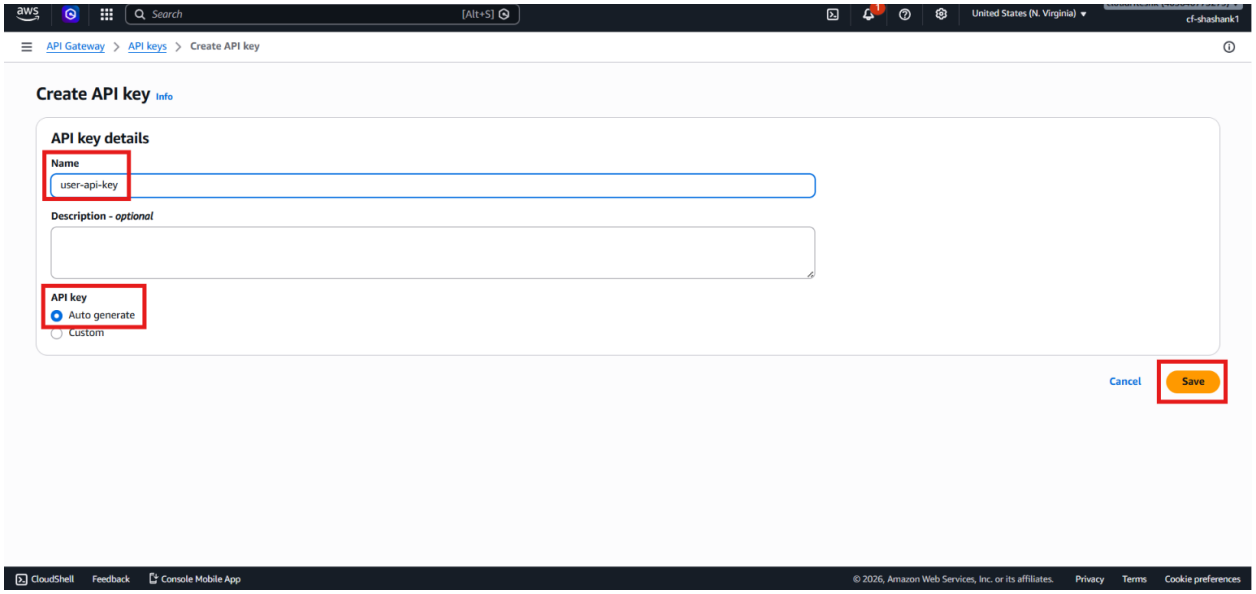
Step 2 – Create an API Key

- Open **API Gateway**.
- Click **API keys**.

- Click **Create API key**.



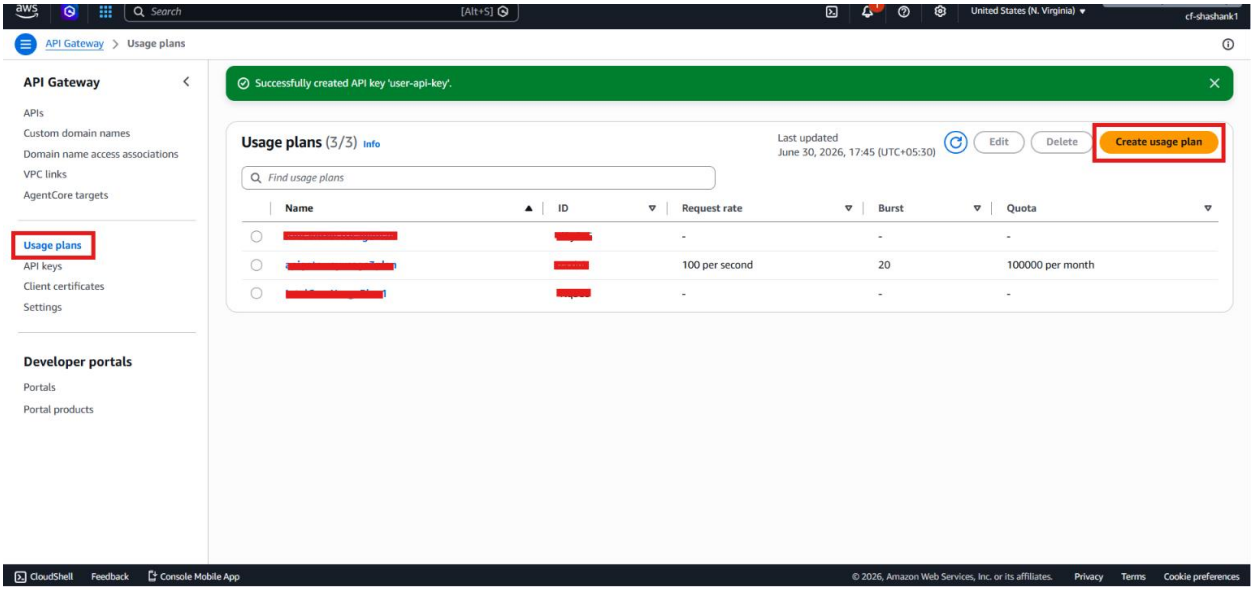
- Enter the key name: user-api-key
- Choose **Auto Generate**.
- Click **Save**.



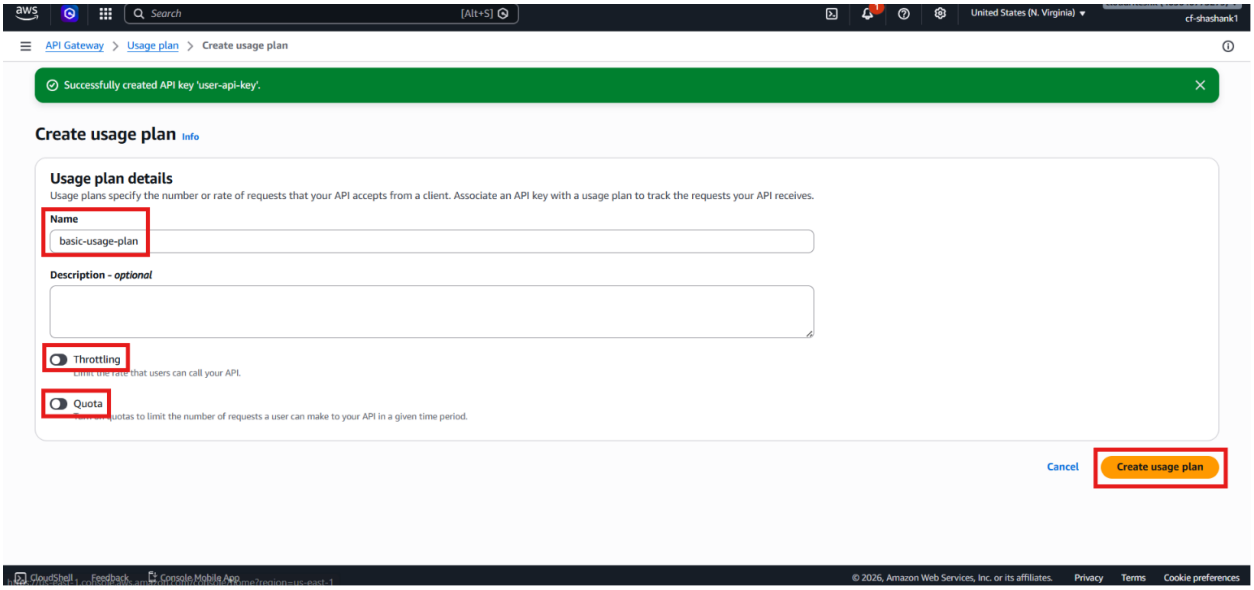
- Copy the generated key and keep it for later use in **Postman**.

Step 3 – Create a Usage Plan

- Open **API Gateway**.
- Click **Usage plans**.
- Click **Create**.

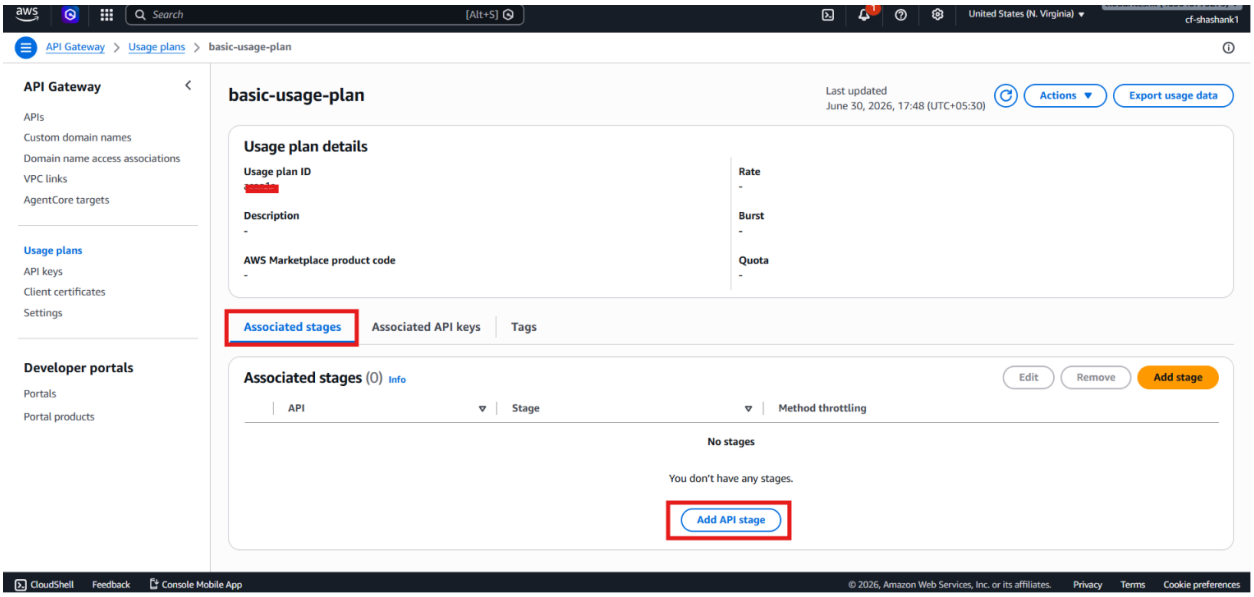


- Enter the usage plan name: basic-usage-plan
- Keep **Throttling** turned off.
- Keep **Quota** turned off for the initial test.
- Click **Create Usage Plan**.

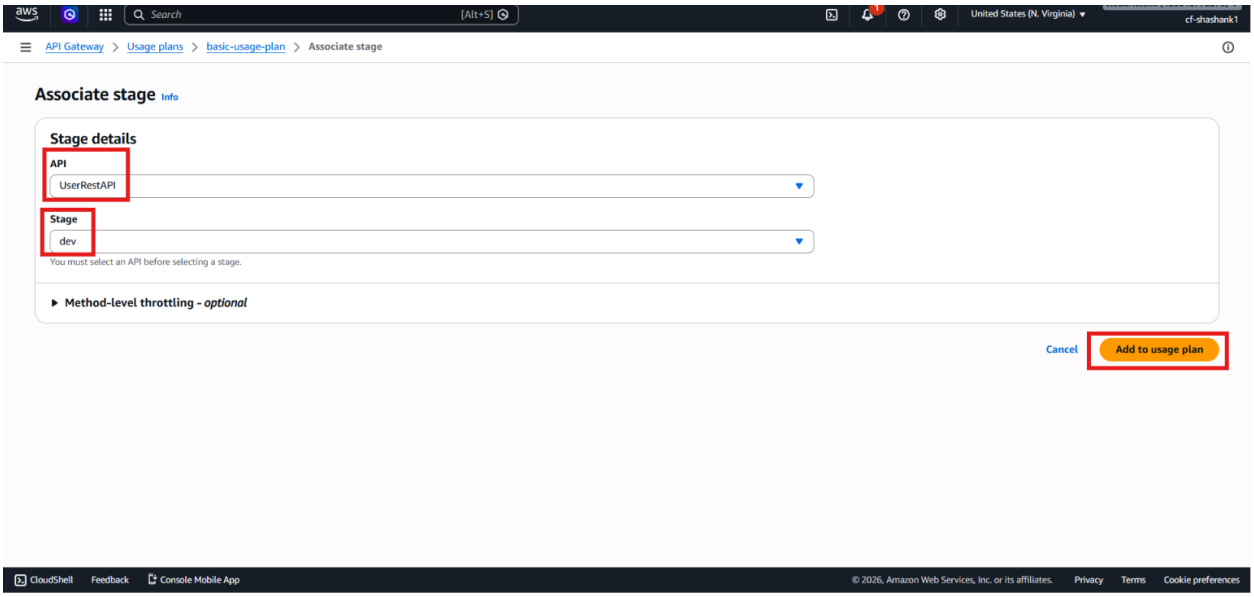


Step 4 – Attach the REST API Stage to the Usage Plan

- Open the **basic-usage-plan**.
- Go to **Associated APIs**.
- Click **Add API**.

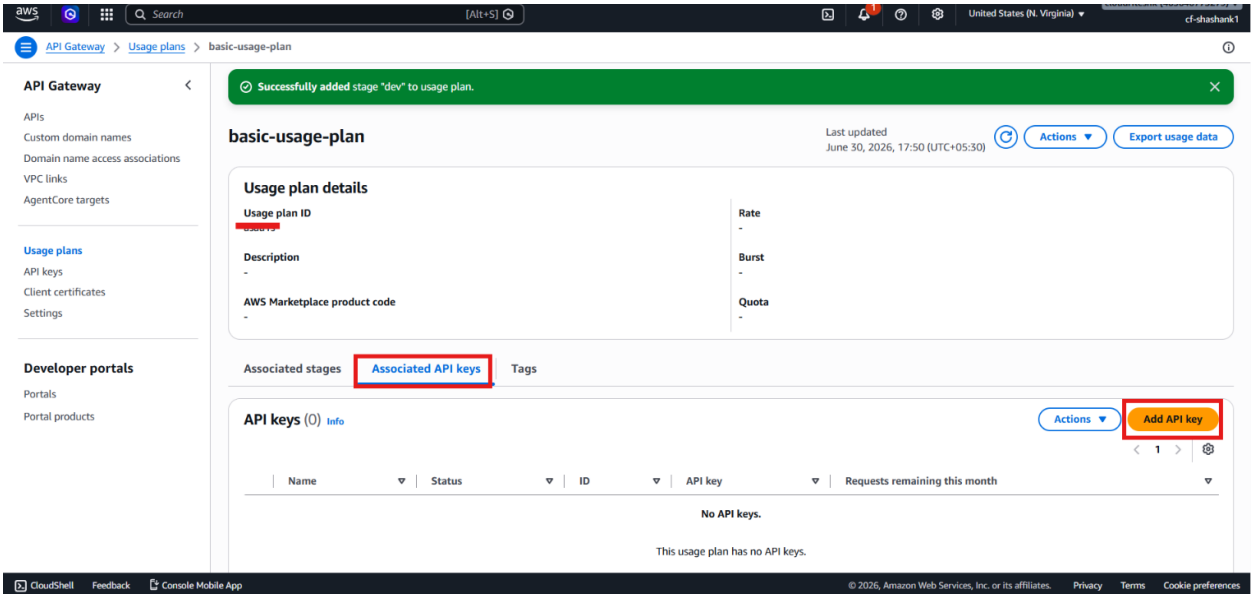


- Select your **REST API**.
- Select the stage **Dev**.
- Save the changes.

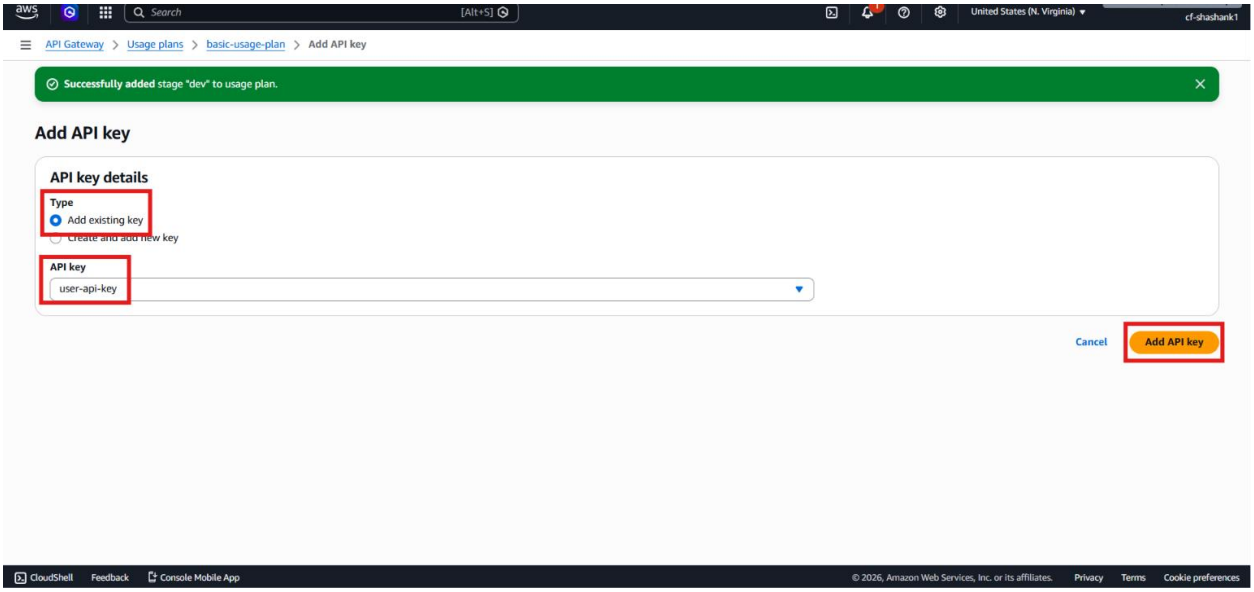


Step 5 – Attach the API Key to the Usage Plan

- Inside the usage plan, go to **Associated API Keys**.
- Click **Add API Key**.

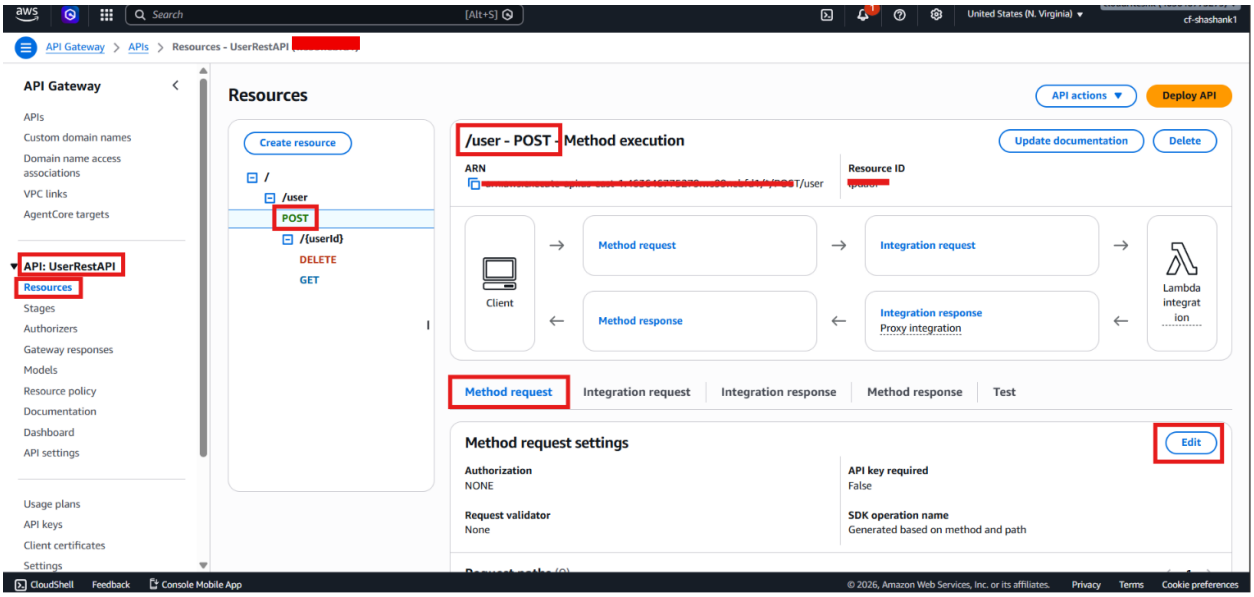


- Select the previously created user-api-key.
- Save the changes.

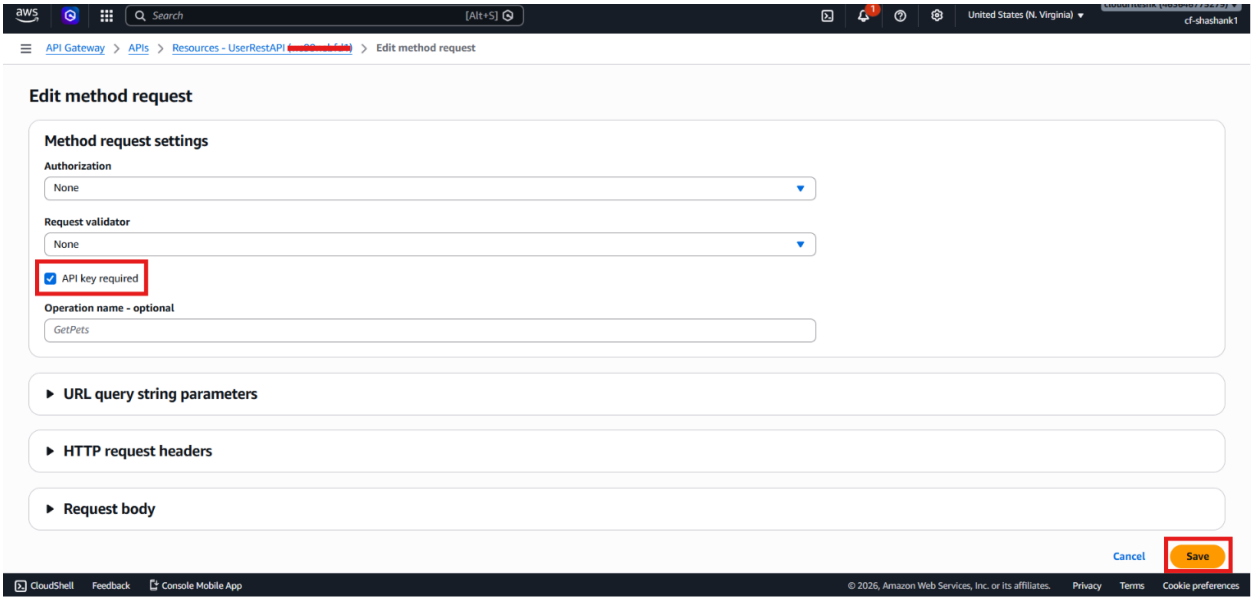


Step 6 – Require the API Key on the Method

- Open your **API → Resources**.
- Select the method you want to protect, such as POST.
- Click **Edit** in **Method Request** tab.

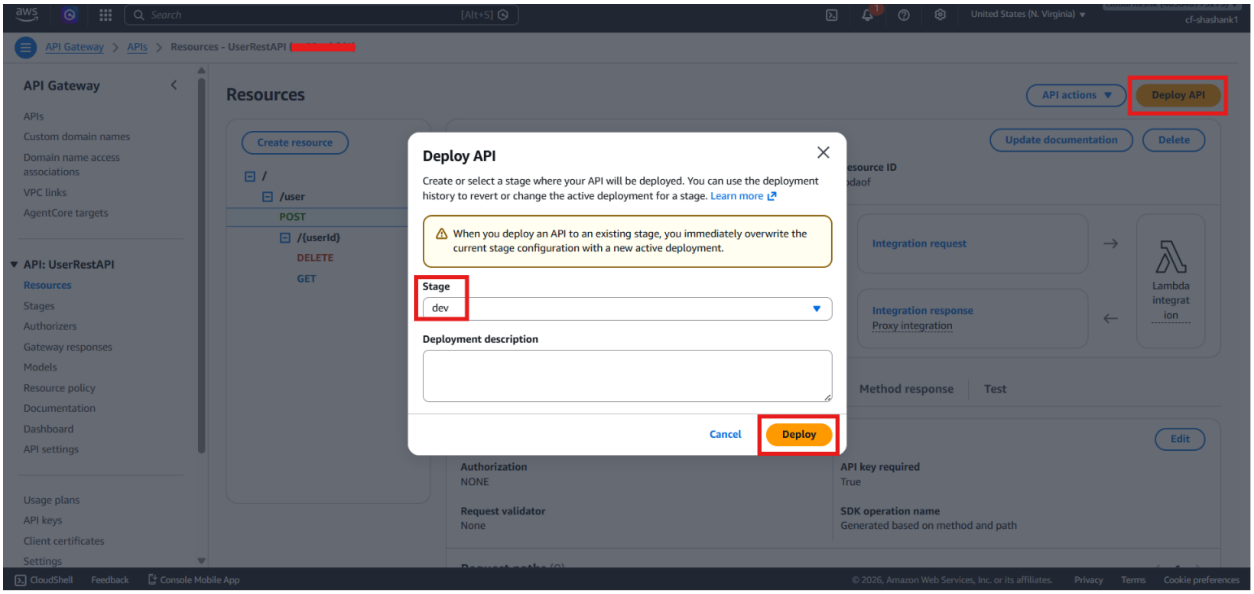


- Set **API Key Required** to **True**.
- Save the method settings.



Step 7 – Deploy the API Again

- Click **Deploy API**.
- Select the existing stage **Dev**.
- Click **Deploy**.



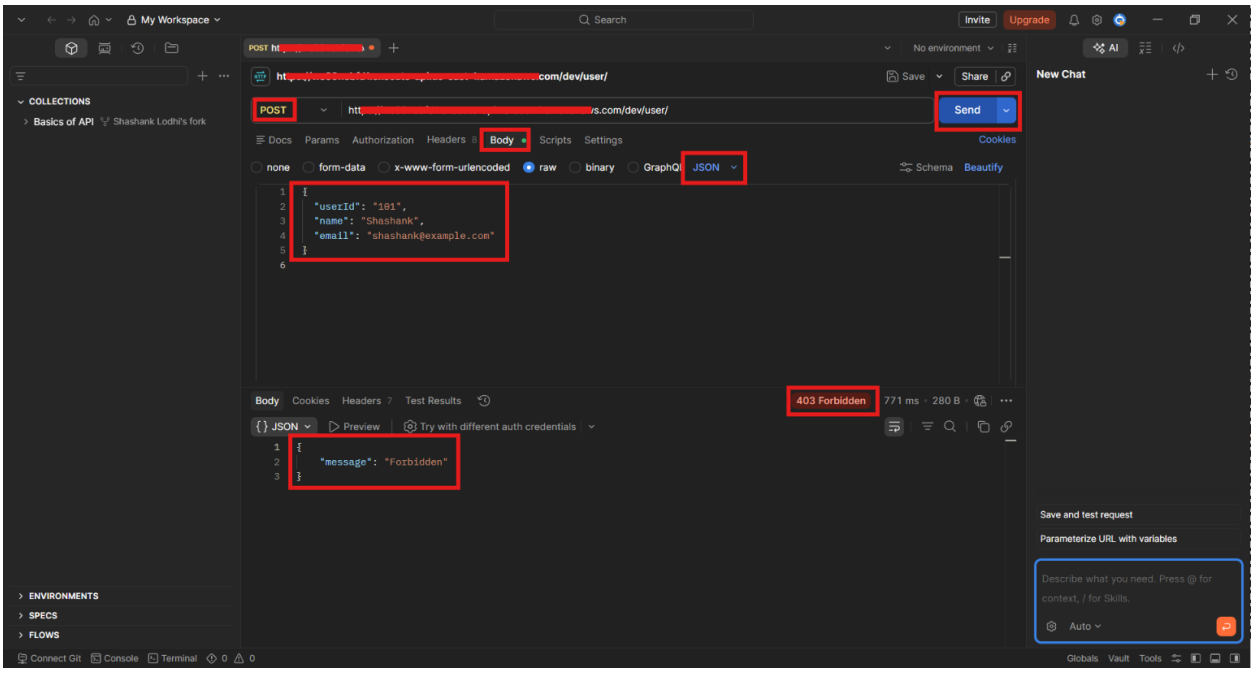
- Wait for the deployment to finish before testing.

Step 8 – Test Without the API Key

- Open **Postman**.
- Create a new **POST** request.
- Paste the invoke URL and append /user.
- Add the request body JSON.

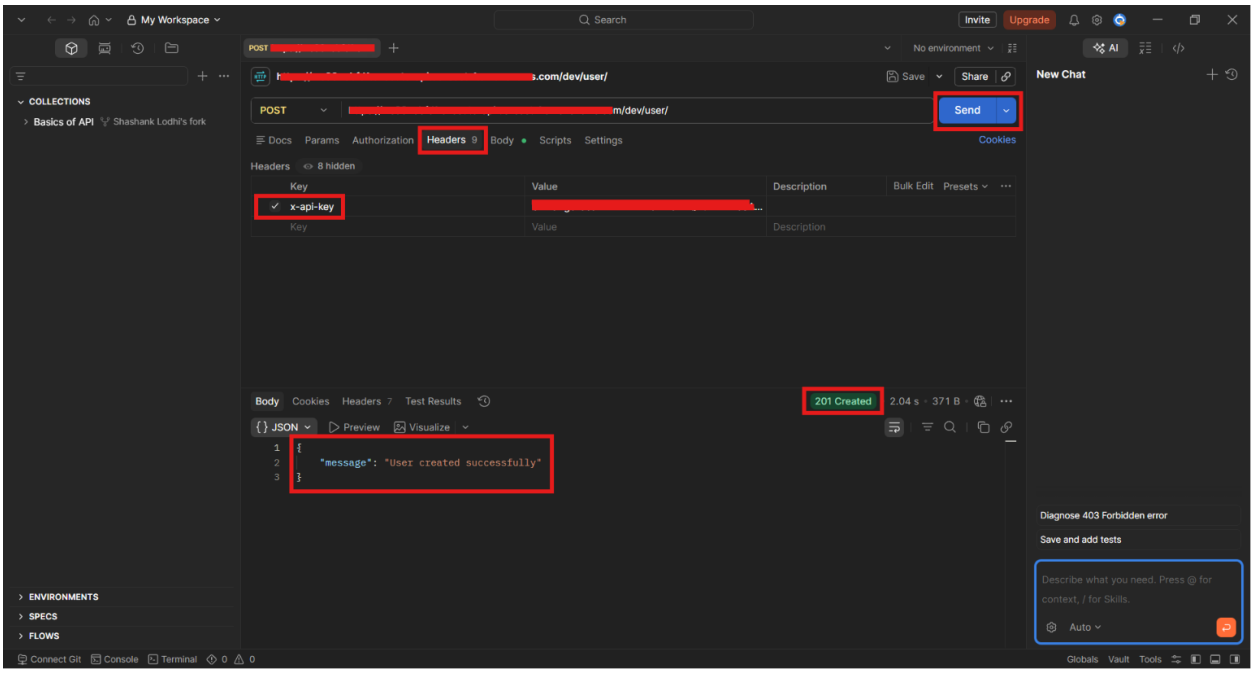
```
{
  "userId": "101",
  "name": "Shashank",
  "email": "shashank@example.com"
}
```

- Send the request without any API key header.
- Confirm that the response is **403 Forbidden**.

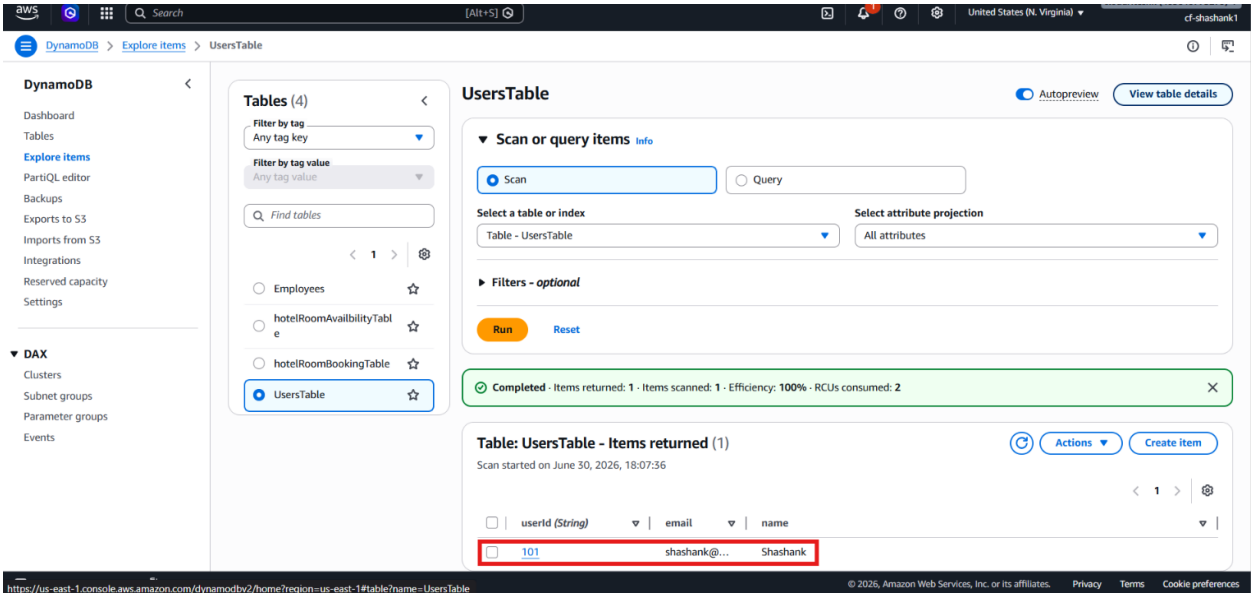


Step 9 – Add the API Key Header in Postman

- In **Postman**, open the **Headers** tab.
- Add the header key: x-api-key
- Paste the generated API key as the header value.
- Send the same POST request again.
- Confirm that the response shows the user was created successfully.

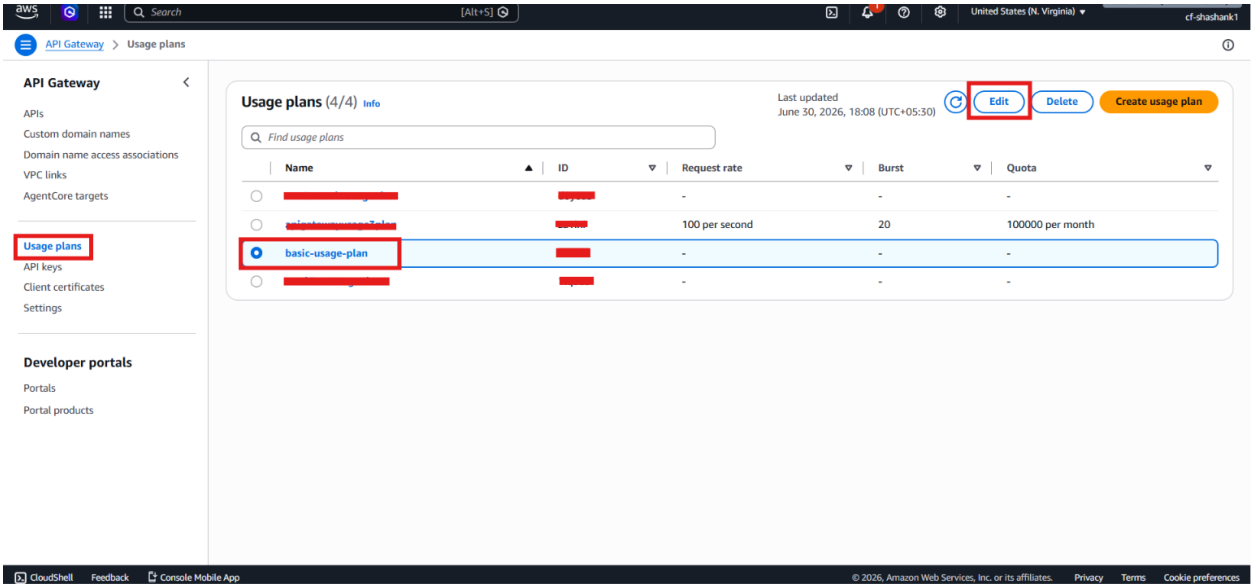


- Open **Amazon DynamoDB** and verify that the record was inserted.

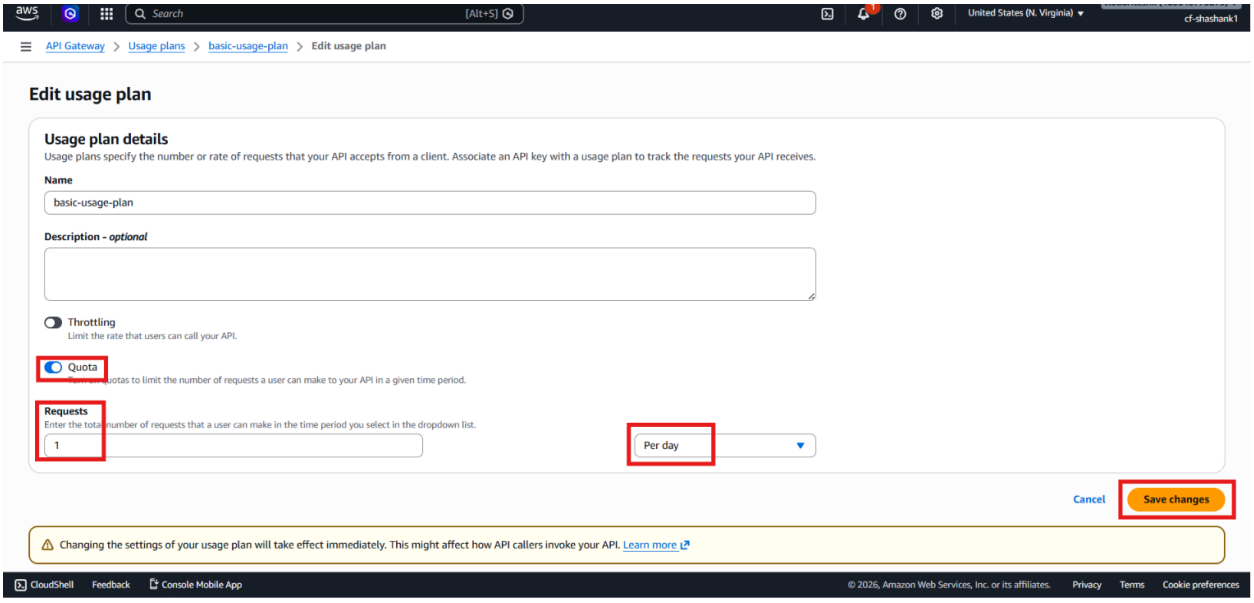


Step 10 – Test the Usage Plan Quota

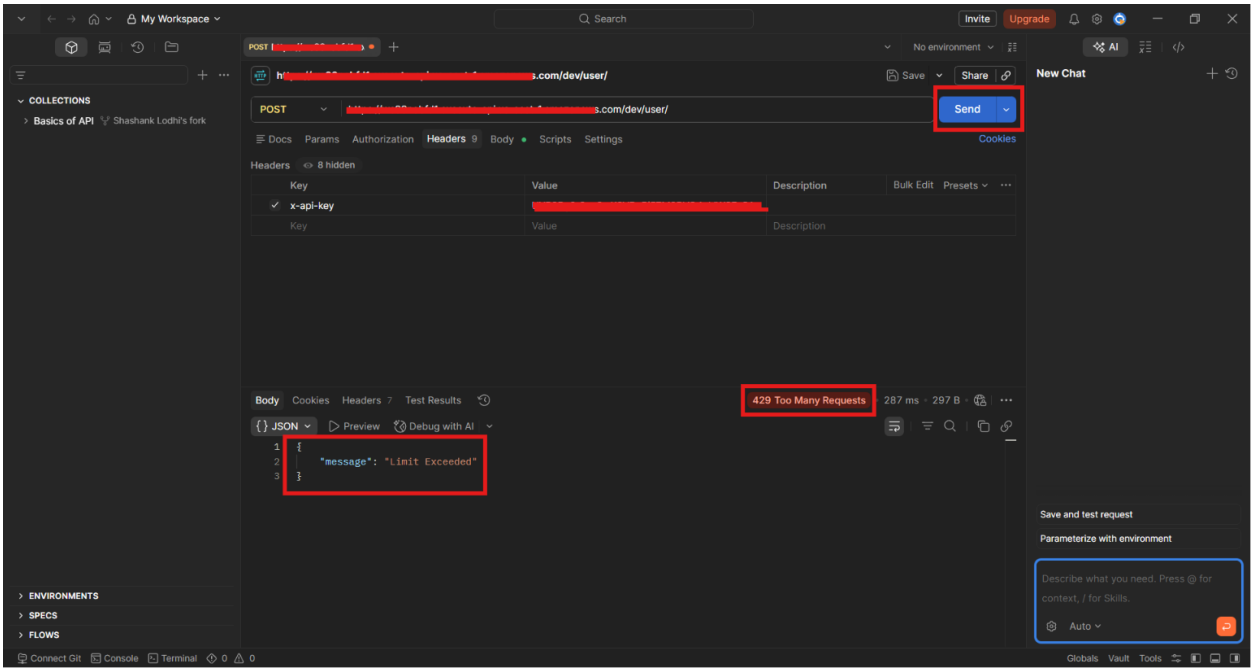
- Return to the usage plan.
- Click **Edit usage plan**.



- Enable **Quota**.
- Set the quota to **1 request per day**.
- Save the changes.



- Deploy the API again.
- Send another request from **Postman**.
- Confirm that the response becomes **429 Too Many Requests** or **message limit exceeded**.



Verification

- The API key was created successfully.
- The usage plan was created successfully.
- The REST API stage **Dev** was attached to the usage plan.
- The API key was attached to the usage plan.
- The POST method was set to require an API key.
- Requests without x-api-key returned **403 Forbidden**.

- Requests with the valid API key reached **Lambda** successfully.
- The quota test returned **429 Too Many Requests** after the limit was reached.